

4COSCOO2W MATHEMATICS FOR COMPUTING

Week 4 Seminar Tasks

Relations and Functions

READING

Lecture 3 Notes (available on Blackboard)

Chapters 16-20. *Croft, T and Davison R (2016) Foundation maths, 6th ed. Harlow: Pearson.*

Chapter 6,7. *Grossman, P., 2017. Discrete mathematics for computing. Bloomsbury Publishing.*

TASK 1. FUNCTIONS VS RELATIONS

See Lectures 3 Notes -Slides 11-19

Let $A = \{\text{Mary, John, Peter, Chris}\}$ and $B = \{\text{Accountant, Lawyer, Programmer, Lecturer}\}$.

Which of the following relations is a function with *domain A* and *co-domain B*:

1. $\{ \langle \text{Mary, Accountant} \rangle, \langle \text{John, Lawyer} \rangle, \langle \text{Peter, Programmer} \rangle, \langle \text{Chris, Lecturer} \rangle \}$
2. $\{ \langle \text{Mary, Accountant} \rangle, \langle \text{John, Lawyer} \rangle, \langle \text{Peter, Programmer} \rangle, \langle \text{Chris, Programmer} \rangle \}$
3. $\{ \langle \text{Mary, Accountant} \rangle, \langle \text{John, Lawyer} \rangle, \langle \text{Peter, Programmer} \rangle \}$
4. $\{ \langle \text{Mary, Programmer} \rangle, \langle \text{John, Programmer} \rangle, \langle \text{Peter, Programmer} \rangle, \langle \text{Chris, Programmer} \rangle \}$
5. $\{ \langle \text{Mary, Accountant} \rangle, \langle \text{John, Lawyer} \rangle, \langle \text{Peter, Lecturer} \rangle, \langle \text{Peter, Programmer} \rangle, \langle \text{Chris, Programmer} \rangle \}$

TASK 2. ANALYSIS OF A FUNCTION

See Lectures 3 Notes -Slides 20,30-31

Given the function:

$$h(x) = \frac{1}{2}x - \frac{3}{4}$$

1. Identify the type of the function.
2. Evaluate the output for the input values: -2, -1, 0, 1, 2. Form the *ordered pairs of x-y values*.
3. Create a *table of x-y values*.
4. Find the points at which the function intersects the x-axis and y-axis.
5. Draw a graph of the function on paper using the information you got from the previous tasks.
6. Plot the function using [[Desmos graphing calculator](#)] and check if you got the same function.

TASK 3. SLOPE OF A LINE

See Lectures 3 Notes -Slides 26-29

Find the equation of the straight line that passes through the points (6, -1) and (-2,7) and sketch it by hand. Check your plot on [[Desmos graphing calculator](#)].

TASK 4. TYPES OF FUNCTIONS

See Lectures 3 Notes -Slides 20-23

Consider the 5 standard functions below:

1. $f(x)=e^x$
2. $g(x)=1/x$
3. $h(x)=x^2$
4. $u(x)=x$
5. $v(x)=\ln(x)$

Plot the curves of the above functions on the same graph on [[Desmos graphing calculator](#)]. Identify the curves of each function on the graph.

TASK 5. THE INTERSECTION OF TWO FUNCTIONS

See Lectures 3 Notes -Slides 32-35

Given the functions p and q:

$$p(x)=x^2-4x-5$$

$$q(x)=x+1$$

Find the intersection points of these functions (using the factorisation method).

Check your answer using [[Desmos graphing calculator](#)].

TASK 6. IDENTIFYING PARALLEL FUNCTIONS

In mathematics and computing, linear functions can be parallel if they share the same slope. For the following pairs of functions, determine whether they are parallel, verify your results using the [[Desmos graphing calculator](#)], and explore their properties.

(a) Identifying Parallelism. Two linear functions are parallel if they have the same slope (the coefficient of x) but different y-intercepts. For the following pairs of functions, determine whether they are parallel:

1. $f(x) = 3x + 2$ and $g(x) = 3x - 4$
2. $f(x) = 4x - 5$ and $g(x) = -4x + 3$
3. $f(x) = 5x + 2$ and $g(x) = \frac{15x+12}{3}$
4. $f(x) = 3(2x - 5) + 4$ and $g(x) = 6x - 2$
5. $f(x) = -\frac{1}{3}x + \frac{7}{3}$ and $g(x) = \frac{2x-6}{6}$
6. $f(x) = -4x + \sqrt{2}$ and $g(x) = -\frac{8x-4\sqrt{2}}{2}$

to be parallel the m should be same

(b) Creating Parallel Functions. Given the linear function $p(x) = -2x + 6$, complete the following:

1. Create two functions that are parallel to $p(x)$ and have y-intercepts of 4 and -7, respectively.
2. Graph these functions using the Desmos graph calculator.
3. Identify the y-axis intercepts of each function and explain why parallel functions, even though they have different y-intercepts, will never intersect each other.

TASK 7. SET THEORY AND FUNCTIONS IN COMPUTING

See Lectures 3 Notes -Slides 14, 16, 24-25

In computing, functions map elements from one set to another. Not all elements in the sets are always involved in the mapping, so it's important to understand the concepts of *domain* (the set of all inputs for which a function is defined) and *co-domain* (the set of all possible outputs).

Consider the following two sets:

- Set $A = \{2, 4, 6, 8, 10\}$
- Set $B = \{1, 3, 5, 7, 9\}$

Let $f: A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} + 1$, where the output must belong to set B . Complete the following tasks:

(a) Function Mapping. For each element in set A , compute the corresponding output in set B using the function $f(x) = \frac{x}{2} + 1$. If the output does not belong to set B , explain why the function is undefined for that input.

(b) Define the Domain and Codomain. Based on your calculations in part (a), identify the domain (the set of inputs for which the function is defined) and the codomain (the set of valid outputs from the function).

(c) Finding the Inverse. Given that the function f is invertible for some of the elements in A and B , find the inverse function $f^{-1}(x)$ for the valid pairs. Then compare the input-output sets of the function and its inverse.

TASK 8. THE INVERSE OF A FUNCTION

See Lectures 3 Notes -Slides 24-25

Given the function:

$$h(x) = \frac{3x - 4}{2}$$

1. Find the inverse of the function $h(x)$.
2. Confirm that the original function and its inverse are reflections across the line $y = x$ using algebraic verification: $h(h^{-1}(x)) = x$ and $h^{-1}(h(x)) = x$.
3. Plot the graphs of $h(x)$, $h^{-1}(x)$, and $y = x$ to visually confirm that h and h^{-1} are reflections of each other across the line $y = x$.

TASK 9. RECURSIVE FUNCTIONS

See Lectures 3 Notes -Slides 36-38

Task 9.1. Explore a sequence $p(n)$ defined for all non-negative integers using the following recursive definition:

Base Case (BC): $p(0) = 2$

Recursive Case (RC): $p(n+1) = p(n)^2 + 1$, for $n \geq 0$.

Calculate the value of $p(4)$.

Note: You may need to use a calculator for this task.

Task 9.2. Let $t(n)$ be a sequence defined for all non-negative integers by the following recursive definition:

Base Case (BC): $t(0) = 1$; $t(1) = 3$

Recursive Case (RC): $t(n) = t(n-2) + t(n-1)$, for $n \geq 2$

Find out the value of $t(7)$.

Task 9.3. Investigate a sequence $f(n)$ that employs the *Factorial function* in its definition for all non-negative integers:

Base Case (BC): $f(0)=1$

Recursive Case (RC): $f(n+1)=f(n)+n!$ for $n \geq 0$

Determine the value of $f(4)$.

TASK 10. MODULAR ARITHMETIC AND FUNCTIONS IN COMPUTING

In computing, modular arithmetic is often used to manage data in circular arrays or queues, where indices wrap around after reaching a certain value. The function $f(x) = (2x + 3) \bmod 7$ represents a method for calculating indices in a circular array of size 7 based on the input x , which could represent the position in a sequence of operations or data.

Using the function $f(x)$, calculate the corresponding indices in the circular array for the following values of x :

- a) $f(4)$
- b) $f(8)$
- c) $f(11)$
- d) $f(15)$
- e) $f(18)$